Model Forge — Plugin Datasheet

Status report — 2026-11-06. Hermes Agent Plugin Hackathon entry. Repository: Gitea charles/model-forge (public GitHub mirror pending for submission). Deadline: 5:00 PM EST, Tuesday 2026-11-11.

1. What It Is

Model Forge is a Hermes Agent plugin family that makes local AI a first-class experience across every modality: chat, vision, OCR, image generation, image editing, video, 3D meshes, music, sound effects, and voice. It finds the user's hardware, shows which models fit, drives the right inference engine (llama.cpp, Ollama, Strata, Project Maya, vLLM / vLLM-Omni, SGLang / SGLang Diffusion, ComfyUI, or any OpenAI-compatible endpoint), pulls the model, and wires it into Hermes as a provider — so every native tool (chat, image_generate, video_generate, vision, TTS) can use it. The one-line pitch: 'Find your GPU, pick a model, and Hermes runs it — no account, no API key, nothing leaves your machine.'

Judges score usefulness, reliability, polish, and creativity, and favor plugins that do more than a prompt or a skill could. Model Forge manages engine lifecycles, hardware-aware model selection, and cross-machine compute targets — runtime behavior no skill can emulate.

2. Architecture Map — Five Packages

Package	Kind	Role
model-forge	general (tools)	6 agent tools: status, browse, target_add, install, activate, generate_media. Capability registry (11 modalities), hardware probe, engine detection, pull jobs.
model-forge-providers	model-provider	Registers 'strata' and 'maya' as first-class Hermes model providers (keyless local endpoints, LM Studio profile pattern).
model-forge-image	backend	Routes image_generate to any OpenAI-compatible images API (broker, vLLM-Omni, SGLang Diffusion). Live model discovery; edit + 4 reference images.
model-forge-video	backend	Routes video_generate to any OpenAI video job API (create → poll → content) via the official OpenAICompatibleVideoGenProvider base.
model-forge-desktop	desktop	Cockpit pane: every compute target with live engine probes and model counts; palette refresh command.

3. Agent Tools

Tool	What it does
model_forge_status	GPUs, VRAM, RAM and engine inventory per compute target; pull-credential booleans.
model_forge_browse	Models by modality with advisory hardware-fit verdicts per target (fits_gpu / fits_memory / needs_gpu / too_big / engine_managed).
model_forge_target_add	Register a compute target: this machine, a LAN LXC, or a rented cloud GPU box. Persisted; Desktop JSON editor feeds the same list.
model_forge_install	Pull a model: Maya HTTP pull (byte progress), Hugging Face download job (resume + sha256 + job_id polling), engine CLI commands for Strata; run-on-target instructions for remote boxes.
model_forge_activate	Connect a running chat/vision model as Hermes' default. Plan-only unless apply=true. Emits media_setup wiring commands for image/video providers.
model_forge_generate_media	3D (image to GLB, submit job then poll), music and SFX through engine job APIs. Honest route probing: a target without the lane says so.

4. Engine Support Matrix

Engine	Fingerprint probe	Modalities	Status
llama.cpp / llama-server (incl. Hermes managed runtime)	GET /props	chat, vision	detect + core Local Models defer
Ollama	GET /api/tags	chat, vision, ocr	detect
Strata (125B MoE on consumer GPUs)	local CLI + /v1/models	chat, vision	detect + strata launch argv
Project Maya (321B-748B MoE, 1-16 GPUs)	GET /api/catalog	chat, vision, ocr	detect + full Models-page API (pull/poll/cancel/delete)
vLLM / vLLM-Omni	GET /version + images-route probe	chat, vision, ocr, image, image_edit	detect
SGLang / SGLang Diffusion	GET /get_model_info + images-route probe	chat, vision, image, image_edit	detect
ComfyUI	GET /system_stats	image, image_edit, video, 3d	detect
Any OpenAI-compatible endpoint	GET /v1/models (never claims engine identity)	chat, vision, ocr, image, video	generic detector + capability probes

Route probing reads /openapi.json first (authoritative for FastAPI apps), then falls back to GET and empty-POST probes. This matters in practice: the weights broker returns 404 on both wrong-method GETs and empty POSTs for routes that exist.

5. Compute Targets — 'Where Does the GPU Live'

A compute target is a machine with GPUs: this machine, an LXC on a Proxmox node, or a rented cloud box. Targets carry an engine URL and declared VRAM/RAM budgets used for fit verdicts. Three ways to manage them: the model_forge_target_add tool, the Desktop Settings > Plugins JSON editor (config_schema targets_json), or the cockpit pane. The reference deployment is 'ai-lxc': an LXC on a Proxmox node running a weights broker on 2x RTX 3090 (48 GB VRAM + 96 GB RAM) — discovered and driven over the LAN from a GPU-less VM.

Pull credentials: optional Hugging Face token (HF_TOKEN) and ModelScope token (MODELSCOPE_API_TOKEN), entered as masked fields in Desktop Settings > Plugins. Stored through Hermes' credential route, read only via the secret scope, and host-scoped in code: a token is only ever attached to its own host family (huggingface.co / modelscope.cn) — unit-tested, including lookalike hosts.

6. Verification Evidence

What	How verified	Result
Unit tests	Registry integrity, fit logic, install/activation planners, credential routing, media routing (stdlib runner)	22/22 pass
Catalog gate	hermes plugins validate + hermes plugins doctor on all 5 packages	clean (security scan safe, no core override)
Live agent session	Fresh hermes chat calling the tools through the tool bridge	status + browse work; LAN target probed live
Image generation	model-forge-image provider to the broker: gen/moss-image and gen/qwen-image-2.1	real PNGs (2.2 MB and 801 KB)
3D generation	model_forge_generate_media: test image to GLB through a broker Hunyuan3D lane	valid glTF 2.0 (2.6 MB)
Model downloads	pull job path: Hugging Face and ModelScope files, resume + progress jobs	both completed, byte-verified

What	How verified	Result
Hermes update compat	re-ran the full battery after Hermes v0.22.6 to v0.23.0	all green, no fixes needed

7. Broker Repair (companion work, 2026-11-06)

The reference GPU box's image lanes were broken by three stacked problems, all now fixed and verified: (1) workflows were wired for text encoders that never landed — qwen3.5_9b_bf16 was merged on-box from shards already present (zero download), qwen3vl_8b_bf16 pulled from a public mirror (17.5 GB, sha-verified); (2) a corrupt 10 GB UNET file (zero-byte header) — the workflow now uses the intact int8 variant of the same model; (3) a ComfyUI master auto-update regression in the comfy_kitchen flash decode kernel — routed through the wrapper's torch reference fallback and persisted across boot resets. Full lessons recorded in the weights-broker-packer skill.

8. Timeline and Remaining Work

Milestone	Status
D0 Scaffold: registry, hardware probe, first tools	done
D1 Strata + Maya adapters, keyless providers	done
D2 Compute targets, target-aware tools	done
D3 Multi-engine adapters v2, install/activate, credentials	done
D4 Image/video provider packages, pull jobs	done
D5 Media tool (3D/music/SFX), cockpit pane, TTS provider	done
D6 Full storyboard rehearsal + demo video (90 s max)	next
D7 Catalog PR (SHA-pinned) + form + video post	due Nov 11, 5 PM EST

Remaining risks: the demo video needs a recording pass (screen capture of the storyboard chain on real hardware); the catalog submission needs a public GitHub mirror of the Gitea repo; and one visual check of the cockpit pane in the desktop app is outstanding (its engine probes may be limited by the app's cross-origin policy — the pane degrades to a hint row if so).

9. Design Principles

1) Extend Hermes only through public surfaces (provider hooks, register_* APIs, Desktop SDK) — catalog-grade from day one. 2) Never duplicate core: GGUF chat models defer to Hermes' built-in Local Models catalog, which already prices models per machine. 3) Honest capability claims: a generic endpoint is never mislabeled as a named engine; a missing lane gets a clear answer, not a broken call; fit verdicts are advisory and the engine's own sizing is authoritative. 4) Secrets only through Hermes' credential route. 5) Every claim in this document is backed by a live run, not by design intent.